

TidyTuesday Week 41

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Wednesday 7th October, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we throw two unrelated economic series into a blender and act shocked when something comes out. This week was a short one, just two glorious draws, and both of them delivered that classic arc of hope-followed-by-crushing-statistical-honesty that we've all come to cherish.

Monday the sixth gave us a real pairing for the ages: residential natural gas prices in the Middle Atlantic versus the percent of value of syndicated commercial loans at U.S. branches of foreign banks, a series so loved it was eventually discontinued. The raw regression came out swinging – R-squared of 0.44, a p-value of 0.001, and a confident negative slope suggesting that as the gas gets pricier, the foreign-bank loan-syndication percentage slumps. For one shining moment I prepared to write a think-piece connecting Mid-Atlantic heating bills to global banking behavior. Then I detrended it, and the whole thing evaporated into nothing – R-squared of 0.02, a p-value of 0.56, and a slope that could not be less distinguishable from zero if it tried. Textbook trend mirage: two things that drifted along over 2012-2017 and got mistaken for soulmates. Deeply moving. Deleted the think-piece.

Tuesday the seventh brought bank equity-to-assets ratios in the South Atlantic (also discontinued, we only pair with the dearly departed apparently) against the US-dollar-to-SDR exchange rate. The raw result was statistically significant again – p-value of 0.003 – but with an R-squared of a whopping 0.067, meaning it "explained" roughly seven percent of anything, which is the statistical equivalent of showing up to a potluck with a single ice cube. And of course, once detrended, it too lay down and died: p-value of 0.46, slope flipping sign just to spite me. Another relationship that existed only because both lines were vaguely sloped and lonely.

So the scoreboard this week reads two raw "hits," zero survivors after detrending, which is honestly the house special. As always, the reminder: these series are paired at random, there is no theory, no causality, and no reason on Earth to believe natural gas prices talk to foreign bank loans. Spurious correlation isn't the bug here, it's the entire point of the exhibit. Tune in next week when two more innocent datasets are forced into an arranged marriage they never asked for.

2 Date: 2026-10-06

Series ID: PCU2212102212101122

This series is titled Producer Price Index by Industry: Natural Gas Distribution: Residential Natural Gas for Middle Atlantic Census Division and has a frequency of Monthly. The units are Index Jun 2012=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-06-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: EFAOXFBNQ

This series is titled Percent of Value of Loans Made Under Participation or Syndication for All Commercial and Industry Loans, Other Risk (Acceptable), U.S. Branches and Agencies of Foreign Banks (DISCONTINUED) and has a frequency of Quarterly, 2nd Month's 1st Full Week. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-07-01 and the observation end date is 2017-04-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EFAOXFBNQ	R-squared:	0.444
Model:	OLS	Adj. R-squared:	0.413
Method:	Least Squares	F-statistic:	14.37
Date:	Tue, 06 Oct 2026	Prob (F-statistic):	0.00134
Time:	17:29:15	Log-Likelihood:	-69.835
No. Observations:	20	AIC:	143.7
Df Residuals:	18	BIC:	145.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	104.3514	21.206	4.921	0.000	59.798	148.904
value_fred_PCU2212102212101122	-0.8266	0.218	-3.790	0.001	-1.285	-0.368

Omnibus:	2.921	Durbin-Watson:	1.600
Prob(Omnibus):	0.232	Jarque-Bera (JB):	1.265
Skew:	0.146	Prob(JB):	0.531
Kurtosis:	1.803	Cond. No.	1.10e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

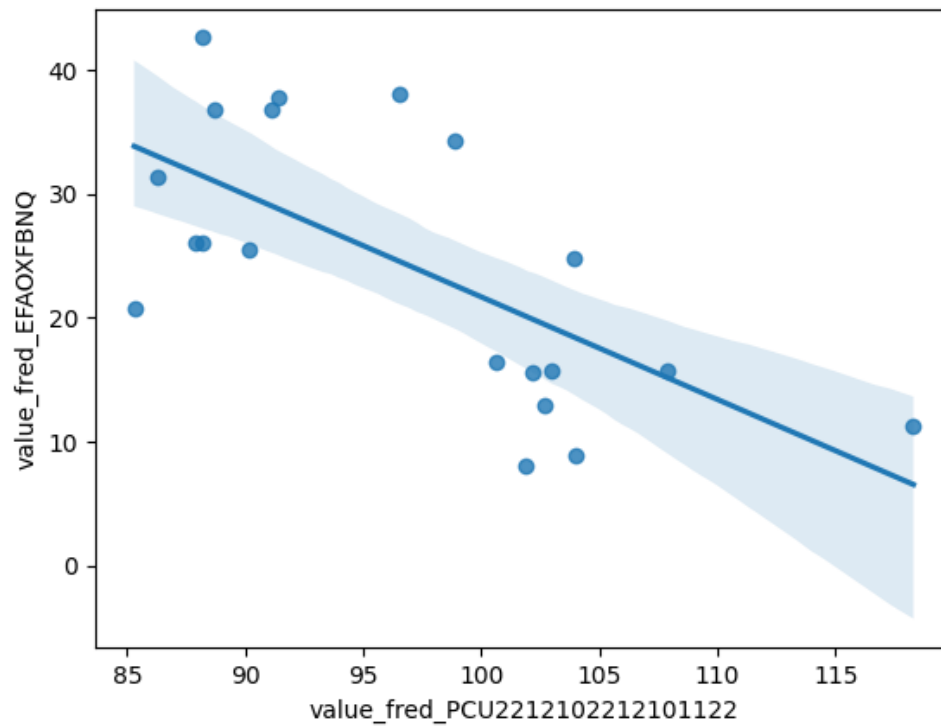


Figure 1: Raw Regression Plot for 2026-10-06

[2] The condition number is large, $1.1\text{e}+03$. This might indicate that there are strong multicollinearity or other numerical problems.

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EFAOXFBNQ_detrended	R-squared:	0.019
Model:	OLS	Adj. R-squared:	-0.035
Method:	Least Squares	F-statistic:	0.3538
Date:	Tue, 06 Oct 2026	Prob (F-statistic):	0.559
Time:	17:29:15	Log-Likelihood:	-69.114
No. Observations:	20	AIC:	142.2
Df Residuals:	18	BIC:	144.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.379e-12	1.807	-2.42e-12	1.000	-3.796	3.796
value_fred_PCU2212102212101122_detrended	-0.2983	0.502	-0.595	0.559	-1.352	0.755

Omnibus:	0.524	Durbin-Watson:	1.639
Prob(Omnibus):	0.770	Jarque-Bera (JB):	0.580
Skew:	0.060	Prob(JB):	0.748
Kurtosis:	2.175	Cond. No.	3.60

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

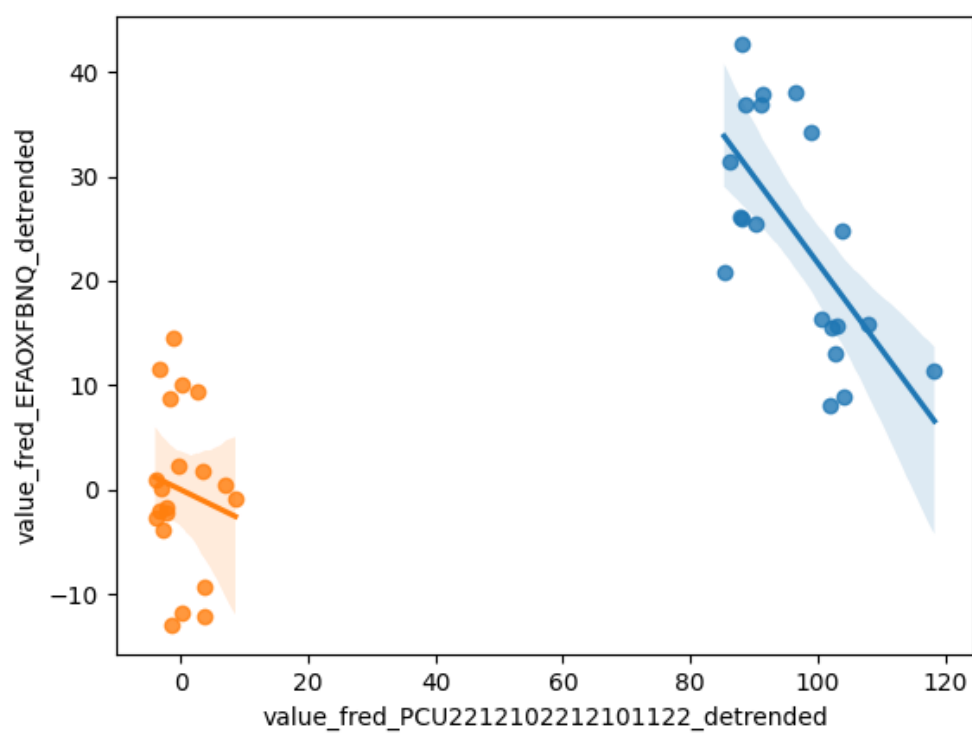


Figure 2: Detrended Regression Plot for 2026-10-06

3 Date: 2026-10-07

Series ID: EQTA35

This series is titled Total Equity to Total Assets, Banks with Total Assets from \$1B to \$10B, South Atlantic Census Division (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 2020-07-01. The popularity of this series is 1.

Series ID: CCUSSP01IFM650N

This series is titled Currency Conversions: US Dollar Exchange Rate: Spot, End of Period: USD: National Currency for SDR and has a frequency of Monthly. The units are Special Drawing Rights and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1975-01-01 and the observation end date is 2023-12-01. The popularity of this series is 13.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CCUSSP01IFM650N	R-squared:	0.067
Model:	OLS	Adj. R-squared:	0.060
Method:	Least Squares	F-statistic:	9.266
Date:	Wed, 07 Oct 2026	Prob (F-statistic):	0.00283
Time:	18:02:18	Log-Likelihood:	133.95
No. Observations:	131	AIC:	-263.9
Df Residuals:	129	BIC:	-258.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.2811	0.049	26.190	0.000	1.184	1.378
value_fred_EQTA35	0.0142	0.005	3.044	0.003	0.005	0.023

Omnibus:	3.573	Durbin-Watson:	0.175
Prob(Omnibus):	0.168	Jarque-Bera (JB):	2.476
Skew:	0.161	Prob(JB):	0.290
Kurtosis:	2.409	Cond. No.	67.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

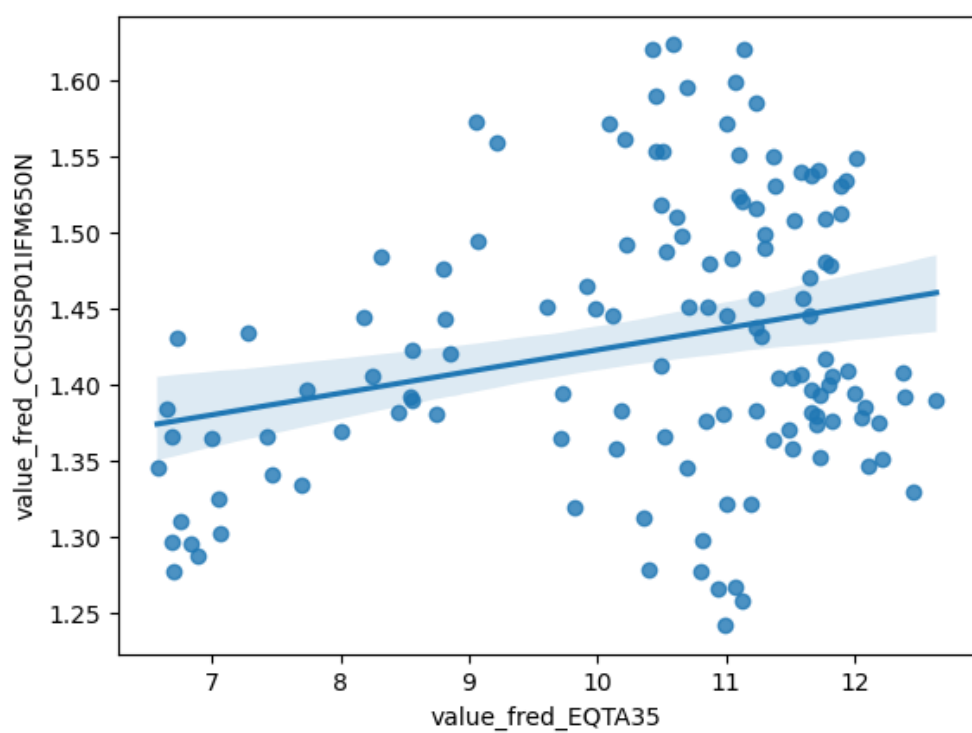


Figure 3: Raw Regression Plot for 2026-10-07

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CCUSP01IFM650N_detrended	R-squared:	0.004
Model:	OLS	Adj. R-squared:	-0.003
Method:	Least Squares	F-statistic:	0.5577
Date:	Wed, 07 Oct 2026	Prob (F-statistic):	0.457
Time:	18:02:18	Log-Likelihood:	139.59
No. Observations:	131	AIC:	-275.2
Df Residuals:	129	BIC:	-269.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
const	3.906e-14	0.007	5.32e-12	1.000	-0.015	0.015
value_fred_EQTA35_detrended	-0.0055	0.007	-0.747	0.457	-0.020	0.009

Omnibus:	10.155	Durbin-Watson:	0.184
Prob(Omnibus):	0.006	Jarque-Bera (JB):	4.169
Skew:	0.123	Prob(JB):	0.124
Kurtosis:	2.162	Cond. No.	1.00

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

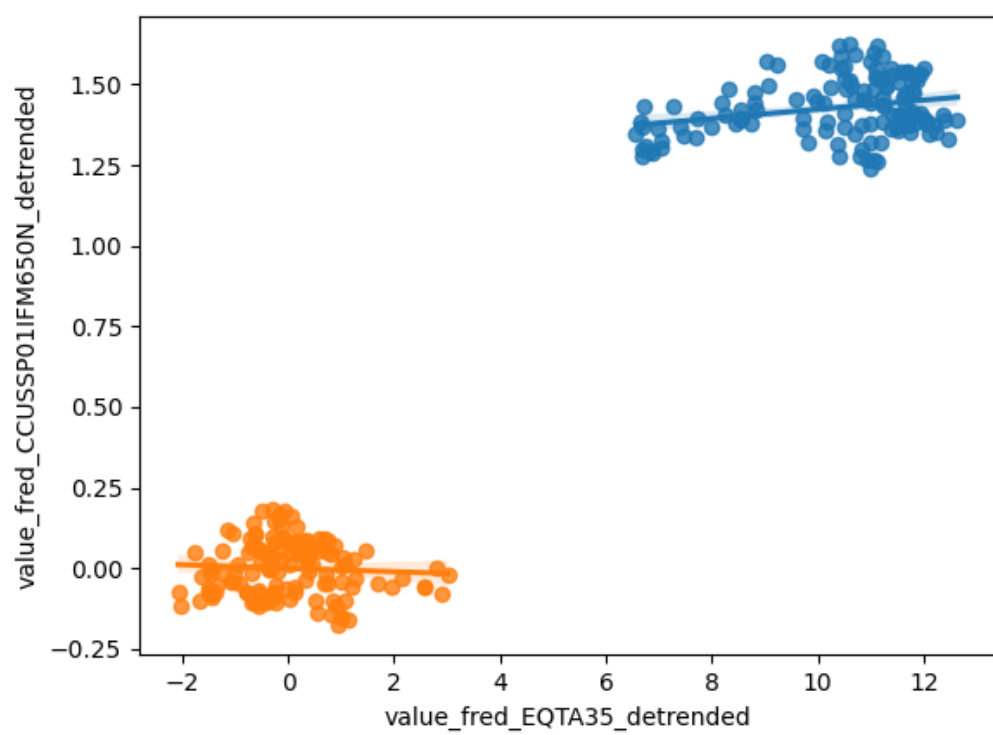


Figure 4: Detrended Regression Plot for 2026-10-07